Technical Report
Project – MPA-ACS (2025/26)

Topic: Detection of sleep apnea from ECG and/or PPG signal

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Description of the algorithm

The algorithm detects sleep apnea events in 1-minute segments of single-lead ECG and PPG signals using a robust voting system.

Signals are first preprocessed. The ECG signal is filtered (5–20 Hz) to isolate QRS energy. R-peaks are identified using a Pan-Tompkins-based integrator followed by a local maximum search (±50ms window) to snap detection to the precise signal peak, ensuring accurate amplitude measurement.

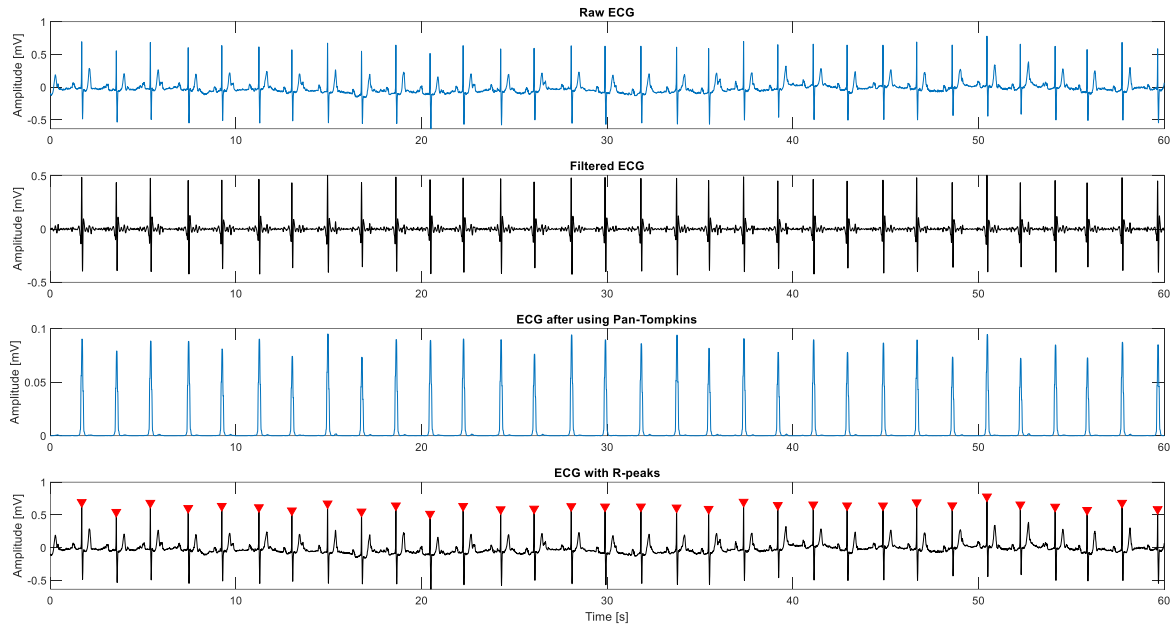

Figure 1: Different stages of ECG signal during preprocessing

The PPG signal is split into two paths: a bandpass filter (0.5–5 Hz) for systolic peak detection and a lowpass filter (<0.5 Hz) to isolate baseline wander.

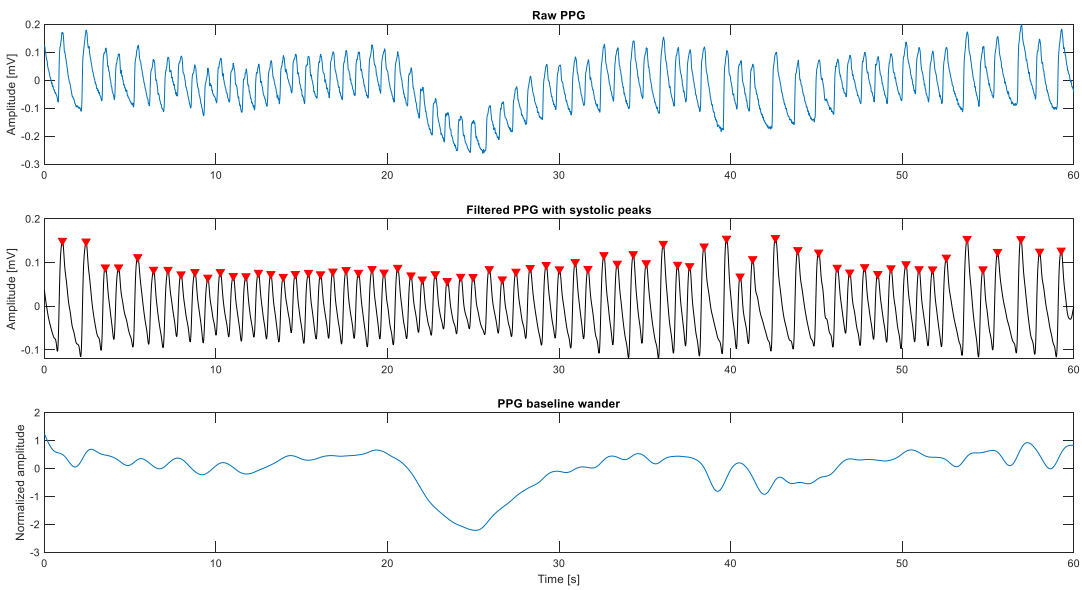

Figure 2: Different stages of PPG signal during preprocessing

Then the algorithm extracts five normalized features to characterize physiological stress:

1. Heart Rate Variability (HRV): Coefficient of variation of R-R intervals [1]

During an apnea event, as oxygen levels drop, the sympathetic nervous system triggers a stress response to wake the brain and restore breathing. This causes the heart rate to swing dramatically, it slows down (bradycardia) during the cessation of breath and speeds up rapidly (tachycardia) upon recovery/gasping. The cyclic bradycardia-tachycardia pushes this variability higher.

R_i is the location (in samples) of the i-th R-peak detected in the ECG signal, and f_s is the sampling frequency (256 Hz). The i-th RR interval, $\Delta t_{RR,i}$, is defined as:

$$\Delta t_{RR,i} = \frac{R_{i+1} - R_i}{f_s}$$

The feature HRV is the ratio of the standard deviation (σ) to the mean (μ) of these intervals:

$$HRV = \frac{\sigma(\Delta t_{RR})}{\mu(\Delta t_{RR})}$$

2. ECG-Derived Respiration (EDR): Standard deviation of R-peak amplitudes [2]

This is a mechanical measure of breathing effort. As the chest expands and contracts, the electrical impedance of the thoracic cavity changes, and the position of the heart relative to the recording electrodes shifts. This physically modulates the amplitude (height) of the R-peaks. Apnea creates higher standard deviation over a 1-minute window.

$A_{R,i}$ is the amplitude of the filtered ECG signal at the i-th R-peak location. First, the amplitudes are normalized by their mean to create a unit-less relative amplitude vector $\hat{A}_{R,i}$:

$$\hat{A}_{R,i} = \frac{A_{R,i}}{\frac{1}{N}\sum_{j=1}^{N} A_{R,j}}$$

The EDR feature is the standard deviation of these normalized values:

$$EDR = \sigma(\hat{A}_R)$$

3. PPG Amplitude Variability: Standard deviation of systolic peak heights [3]

When an apnea event lasts long enough (10-30 seconds), blood oxygen saturation drops. The body's survival reflex is to constrict blood vessels in the extremities (fingers/toes) to shunt oxygen-rich blood to the brain and heart. This causes the pulse wave amplitude in to shrink. The shrink effect creates high variance.

$A_{P,i}$ is the amplitude of the i-th systolic peak in the filtered PPG signal. The normalized amplitude $\hat{A}_{P,i}$ is:

$$\hat{A}_{P,i} = \frac{A_{P,i}}{\mu_{A_P}}$$

The feature PPG amplitude variability is the standard deviation (σ) of the normalized amplitude:

$$PPG_{var} = \sigma(\hat{A}_P)$$

4. Pulse Rate Variability (PRV): Coefficient of variation of PPG peak-to-peak intervals [4]

Acts as a backup to HRV.

Similarly, the P_i is the location of the i-th systolic peak in the PPG signal. The PP interval is:

$$\Delta t_{PP,i} = \frac{P_{i+1} - P_i}{f_s}$$

$$PRV = \frac{\sigma(\Delta t_{PP})}{\mu(\Delta t_{PP})}$$

5. Baseline Wander: Standard deviation of the low-frequency PPG baseline

Detects mechanical struggle and body movement. The patient's diaphragm pulls hard against the closed throat, creating negative pressure in the chest. This suction effect pulls venous blood into the chest, causing the entire baseline of the PPG signal to heave up and down.

The final feature is the standard deviation of the low-frequency PPG baseline:

$$Baseline_{std} = \sigma(B)$$

The classification logic is basically a boolean vote cast for each feature based on optimized thresholds derived from distribution analysis.

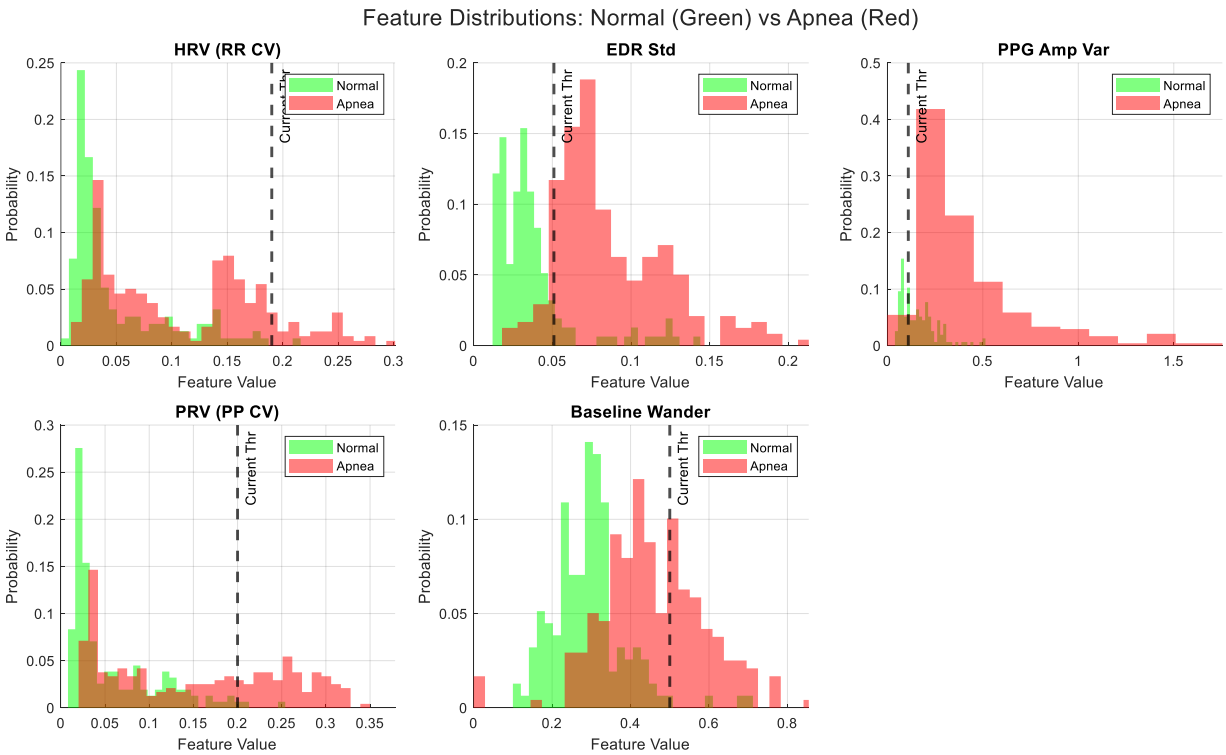

Figure 3: Probability distribution of the features with current set thresholds

To reduce false positives caused by motion artifacts or single-sensor noise, the algorithm employs a consensus voting scheme:

$$Class = \begin{cases} 1(Apnea) & if \sum votes \geq 2 \\ 0(Normal) & otherwise \end{cases}$$

This logic ensures that an event is only flagged if at least two distinct physiological systems (e.g., heart rhythm and chest movement) simultaneously indicate abnormality.

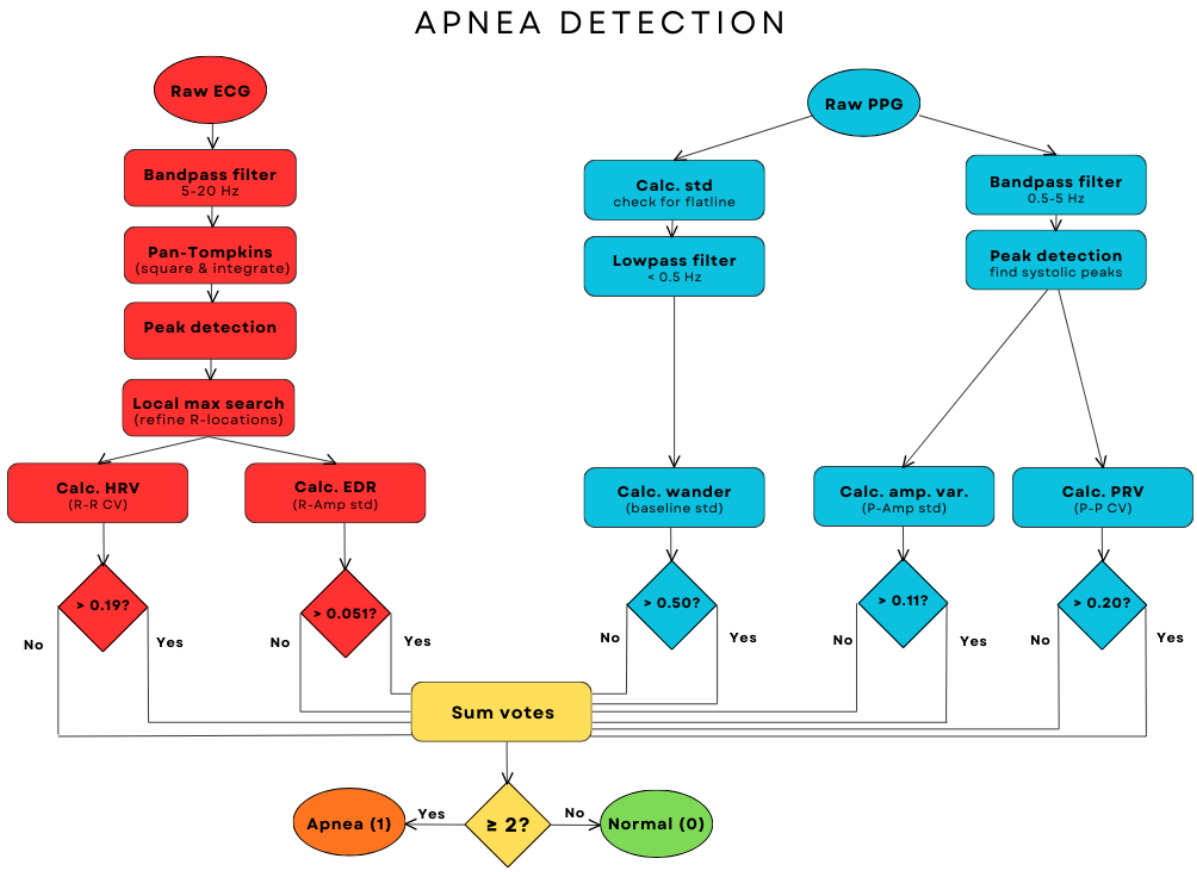

Figure 4: Block diagram of the Apnea Detection Algorithm

Results

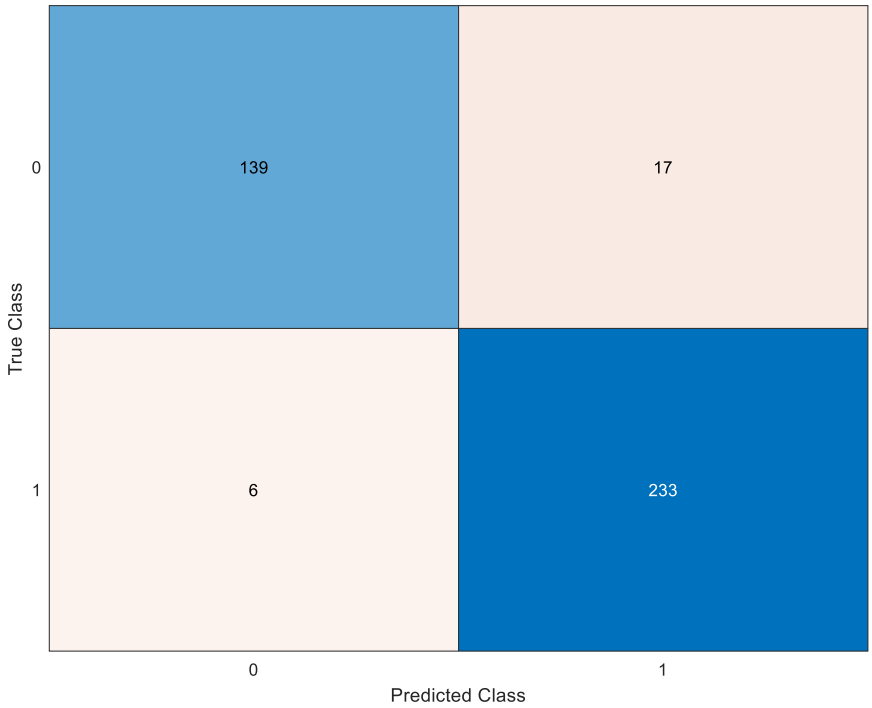

Figure 5: Confusion matrix for training dataset (395 samples)

F1 SCORE = 0.9530

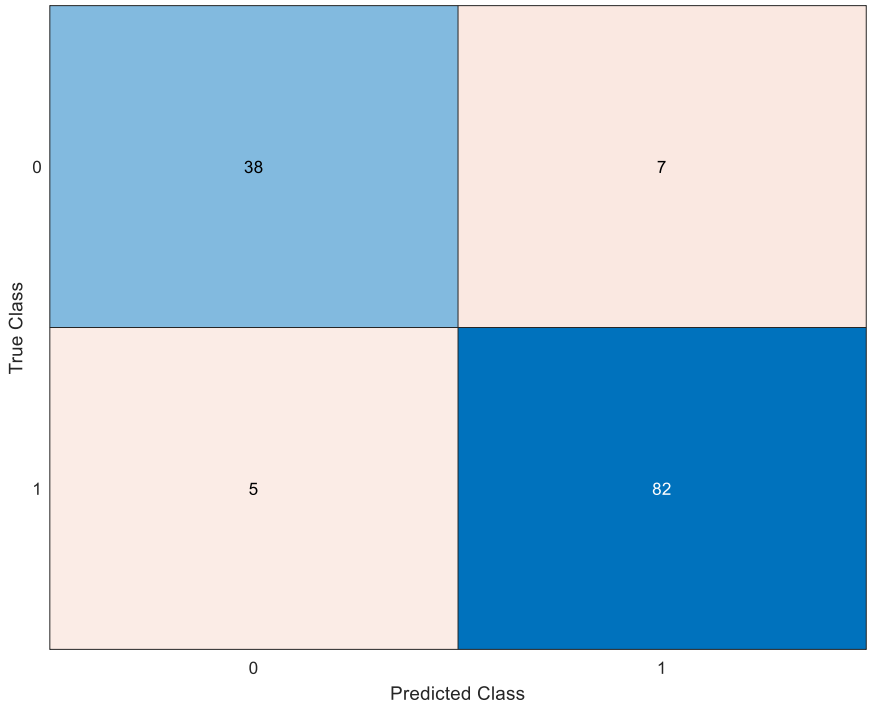

Figure 6: Confusion matrix for validation dataset (132 samples)

F1 SCORE = 0.9318

Discussion and Conclusion

The algorithm achieved an F1 score of 95.30 % on the training set (see Fig.5) and 93.18 % on the validation set (see Fig.6). The minimal drop (2.3 %) indicates that the model is not overfitting. A high validation score demonstrates that the chosen physiological features (HRV, EDR, PPG amp., etc.) are fundamental markers of apnea, rather than random artifacts specific to the training data.

Attempts to implement Decision Trees and Weighted Voting resulted in lower F1 scores. Decision Trees likely overfitted to specific feature hierarchies in the training set that did not apply generally. Weighted Voting likely placed too much trust in single features that occasionally failed due to noise.

Advantages

Unlike Deep Learning models (e.g. CNNs) which act as "black boxes," this algorithm is fully transparent. It can explain exactly why a specific minute was flagged (e.g., "Detected apnea because: EDR > 0.051 AND PPG amp > 0.11"). It uses only basic arithmetic (standard deviation, mean) and standard filters (Butterworth). It requires no heavy matrix operations or GPU acceleration, making it suitable for embedded devices (e.g., smartwatches) with limited battery life. And since this is a rule-based algorithm, it does not require hours of GPU training time. It works immediately once deployed.

Disadvantages

The classification relies on hard-coded thresholds (e.g., 0.19, 0.051). While they work perfectly for the MESA dataset, other datasets might have different noise levels. If we move this code to a device with a different sampling rate, it may be needed to re-tune these numbers manually, because the filter coefficients are calculated relative to fs = 256 Hz.

Although the "Voting" system mitigates noise, a massive whole-body movement could theoretically trigger the Baseline Wander, EDR, and HRV features simultaneously, resulting in a False Positive during normal restless sleep.

Examples

This is an en example of a True Positive case (see Fig. 7). The patient is still (no motion artifact), but their internal systems are struggling. In the top plot (Time ~17s to 35s) we can see the amplitude of the R-peaks decreases and flattens out. This suggests that the patient stopped breathing and was likely trying to breathe against a closed airway. The Baseline (red line) is relatively flat, meaning the patient isn't rolling around. However, when we look at the gray PPG signal they are shrinking and expanding (amplitude modulation). This is the blood vessels constricting due to stress.

This sample proves the value of your multi-modal voting system. A simpler algorithm would have failed here. It successfully detected an apnea event based purely on mechanical (EDR) and vascular (PPG Amp) signs, even though the electrical (HRV) system was silent.

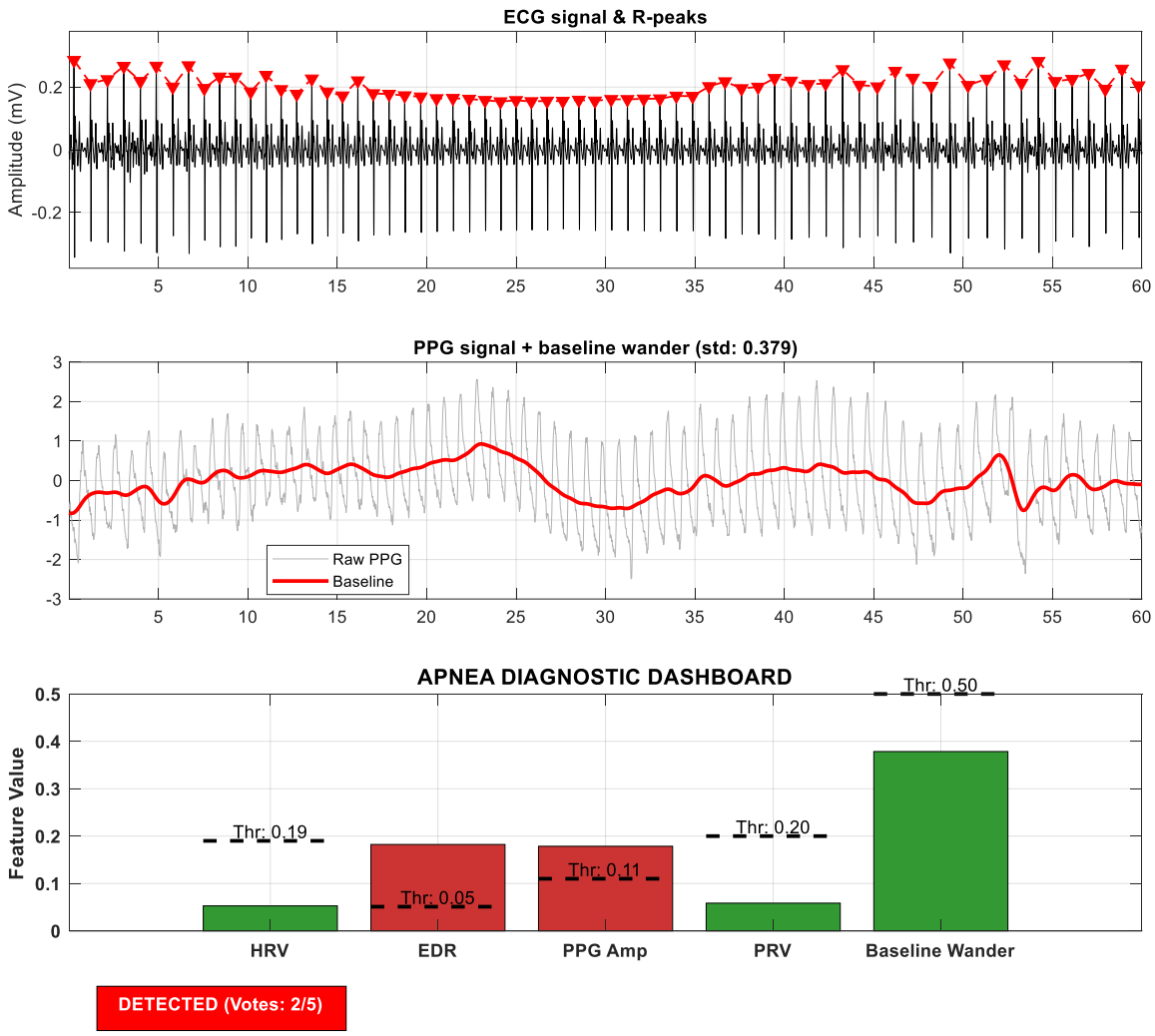

Figure 7: True positive example of apnea detection

This is a textbook example of a motion artifact False Positive (see Fig. 8). The patient is likely Normal (sleeping) but moved their hand or rolled over. The ECG is clean and stable. The R-peaks are perfectly evenly spaced (low HRV) and the amplitude is constant (low EDR). The heart and lungs are behaving perfectly normally. But the PPG signal is chaotic. The massive red waves (baseline wander) are not respiratory (which would be rhythmic), they are mechanical noise. The algorithm got "tricked" because the single event (motion) corrupted the PPG sensor in two different ways, generating two separate votes from the same noise source. This sample highlights the main weakness of a simple "democratic" voting system: correlated errors.

This specific case could be prevented in the future by using sensor logic instead of simple voting: "If Baseline Wander is High (Vote 1), we distrust the PPG Amp vote unless the ECG also agrees." This would have forced the system to look at the ECG, see it was stable, and correctly classify this as Normal.

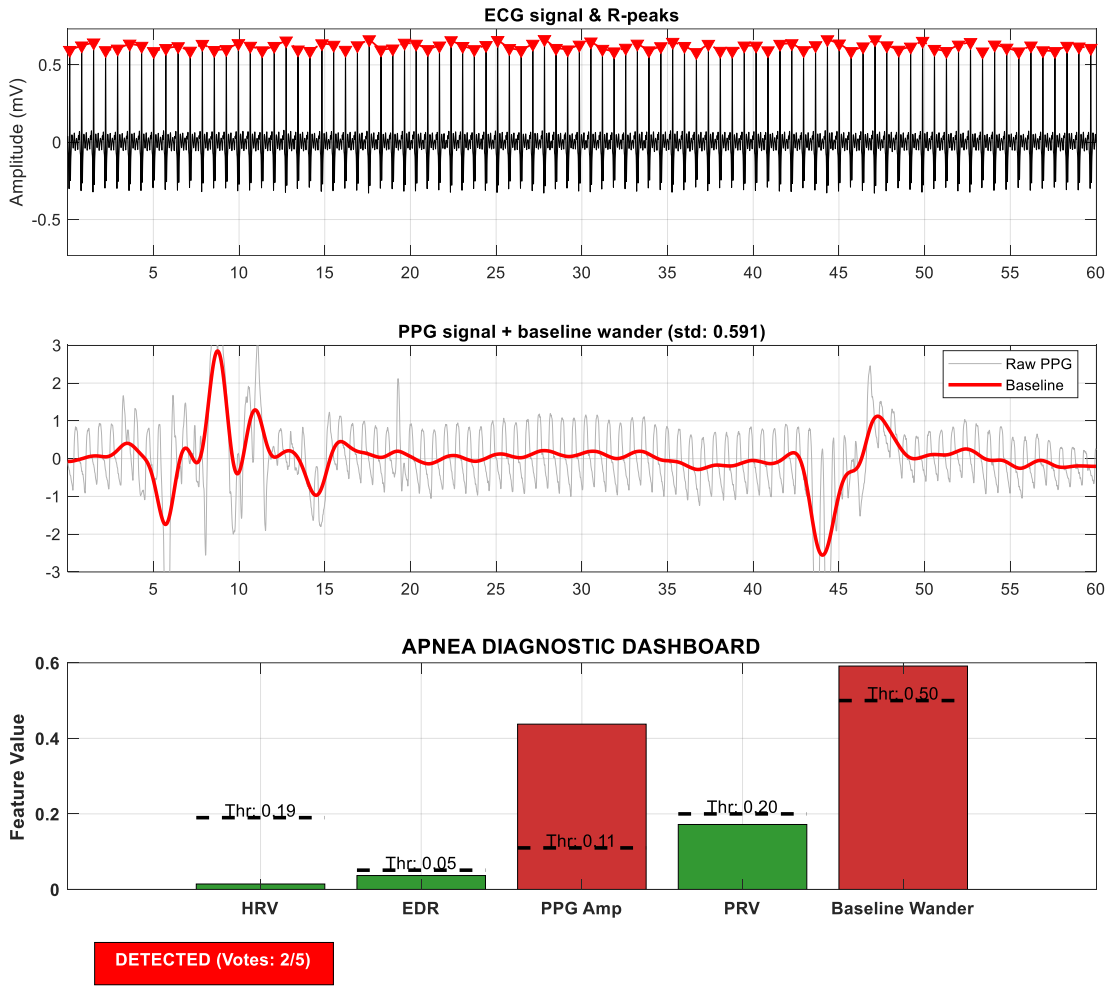

Figure 8: False positive example of apnea detection

References

[1] ROCHE, F. et al. Screening of obstructive sleep apnea syndrome by heart rate variability analysis. Circulation. 1999, 100(13), 1411–1415. DOI: 10.1161/01.cir.100.13.1411.

[2] HELFENBEIN, Eric; FIROOZABADI, Reza; CHIEN, Simon; CARLSON, Eric a BABAEIZADEH, Saeed. Development of three methods for extracting respiration from the surface ECG: A review. Online. *Journal of Electrocardiology*. 2014, vol. 47, no. 6, s. 819-825. ISSN 0022-0736. Dostupné z: https://doi.org/10.1016/j.jelectrocard.2014.07.020.

[3] GIL, Eduardo. Detection of decreases in the amplitude fluctuation of pulse photoplethysmography signal as indication of obstructive sleep apnea syndrome in children. Biomedical Signal Processing and Control. 2008.

[4] LÁZARO, Jesús et al. Pulse rate variability analysis for discrimination of sleep-apnea-related decreases in the amplitude fluctuations of pulse photoplethysmographic signal in children. IEEE Journal of Biomedical and Health Informatics. 2014, 18(1), 240–246. DOI: 10.1109/JBHI.2013.2267096.

[5] Google. Gemini [online]. Mountain View: Google, 2024 [cit. 2025-12-07]. Dostupné z: https://gemini.google.com/

Use of AI

Gemini was used to resolve MATLAB errors that could not be addressed by conventional debugging. It further assisted in the development of the section dedicated to local maxima detection in the ECG preprocessing pipeline, incorporated the baseline wander feature and contributed to the creation of scripts generating figures such as Fig. 7 and Fig. 8.